

Neurons in Motion:

Exploring the Nonlinear Dynamics of Neuronal Firing

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Mathematics

Neurons and Neuronal Firing Basics

- Neurons are a type of neural cell, often considered the building blocks of the nervous system.
- 3 main structures: soma, axon, dendrites.
- Resting membrane potential $\approx -70\text{mV}$ across all species.
- Stimuli initiate **depolarization**: gated membrane channels open allowing Na^+ ions to enter the cell, making the interior more positive.
- Channel openings can generate an **action potential** that propagates from the soma down the axon to the axon terminals.
- After firing, neurons enter **repolarization**: K^+ ions diffuse out of the cell, restoring the resting potential.

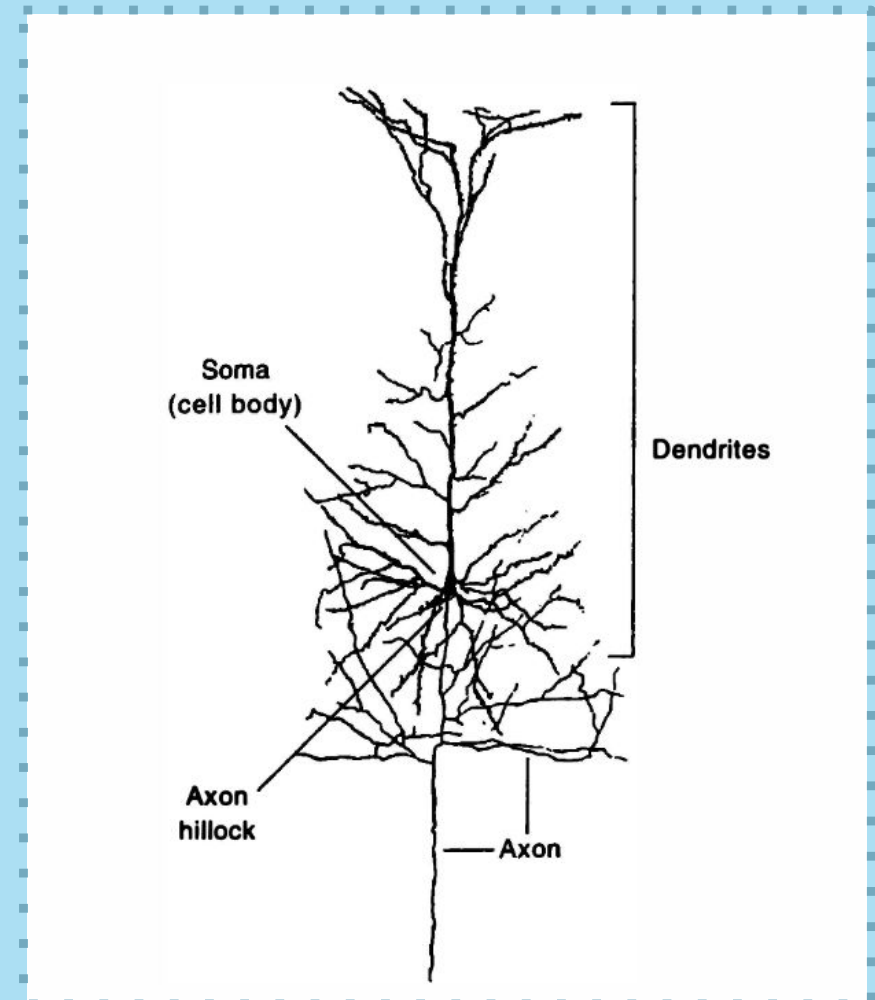
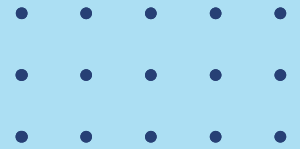


Figure 1: Diagram of a cortical pyramidal cell (Cajal, 1911).

Introduction to Nonlinear Dynamics

- Nonlinear dynamics are ubiquitous across all scientific domains.
- In neuroscience, nonlinear dynamics are particularly prominent, especially in neuron behavior and firing patterns.
- Nonlinearity can pose challenges in the development of mathematical models.
- Nonlinearity can create interesting dynamical behaviors, such as:
 - 1) Memory Effects (Hysteresis, Adaptation, System Forgetting)
 - 2) Noise and Subharmonic Resonance Effects
 - 3) Winner-Take-All Networks
 - 4) Equilibrium Phenomena (Multiple Equilibrium Points, Catastrophe Theory)

Outline For Today



1. Explore the Hodgkin-Huxley equations and Rinzel Hodgkin-Huxley equations, which will serve as our primary models.

2. Observe potential outputs from these models and analyze the implications of these outputs.

3. Examine a case study utilizing the squid axon.

The Hodgkin-Huxley Model

$$C \frac{dV}{dt} = -g_{\text{Na}} m^3 h (V - E_{\text{Na}}) - g_{\text{K}} n^4 (V - E_{\text{K}}) - g_{\text{leak}} (V - E_{\text{leak}}) + I_{\text{input}}$$

$$\frac{dm}{dt} = \frac{1}{\tau_m(V)} (-m + M(V))$$

$$\frac{dh}{dt} = \frac{1}{\tau_h(V)} (-h + H(V))$$

$$\frac{dn}{dt} = \frac{1}{\tau_n(V)} (-n + N(V))$$

Where:

V : Neuronal membrane potential (in mV)
 m : Activation variable for Na⁺ ion channels, $m \in [0,1]$
 h : Inactivation variable for Na⁺ ion channels, $h \in [0,1]$
 n : Activation variable for K⁺ ion channels, $n \in [0,1]$
 I_{input} : Total input current (in μA)
 g_{K} : Conductance of K⁺ ion channels (in mS/cm^2)
 g_{Na} : Conductance of Na⁺ ion channels (in mS/cm^2)
 g_{leak} : Leakage conductance (in mS/cm^2)
 E_{K} : Equilibrium potential of K⁺ ions (in mV)
 E_{Na} : Equilibrium potential of Na⁺ ions (in mV)
 E_{leak} : Leakage equilibrium potential (in mV)
 C : Neuronal membrane capacitance (in $\mu\text{F}/\text{cm}^2$)
 $M(V)$: Activation function for Na⁺ ion channels, $M(V) \in [0,1] \forall V$
 $H(V)$: Inactivation function for Na⁺ ion channels, $H(V) \in [0,1] \forall V$
 $N(V)$: Activation function for K⁺ ion channels, $N(V) \in [0,1] \forall V$
 $\tau_m(V)$: Voltage-dependent time constant for activation of Na⁺ ion channels (in ms)
 $\tau_h(V)$: Voltage-dependent time constant for inactivation of Na⁺ ion channels (in ms)
 $\tau_n(V)$: Voltage-dependent time constant for activation of K⁺ ion channels (in ms)

Auxiliary Equations for the Hodgkin-Huxley Model

Voltage-Dependent Time Constants:

$$\tau_n(V) = \frac{1}{\alpha_n + \beta_n}$$

$$\tau_m(V) = \frac{1}{\alpha_m + \beta_m}$$

$$\tau_h(V) = \frac{1}{\alpha_h + \beta_h}$$

Steady State Values:

$$n_\infty = \frac{\alpha_n}{(\alpha_n + \beta_n)}$$

$$m_\infty = \frac{\alpha_m}{(\alpha_m + \beta_m)}$$

$$h_\infty = \frac{\alpha_h}{(\alpha_h + \beta_h)}$$

Resting State Values:

$$n_0 = \frac{\alpha_{n0}}{\alpha_{n0} + \beta_{n0}}$$

$$m_0 = \frac{\alpha_{m0}}{\alpha_{m0} + \beta_{m0}}$$

$$h_0 = \frac{\alpha_{h0}}{\alpha_{h0} + \beta_{h0}}$$

Voltage-Dependent Transition Functions:

$$N(V) = n_\infty$$

$$M(V) = m_\infty$$

$$H(V) = h_\infty$$

Conductance:

$$g = \frac{I}{(V - E)}$$

Rate Constants:

$$\alpha_n = \frac{0.01(V + 10)}{(\exp(\frac{V+10}{10}) - 1)}$$

$$\beta_n = 0.125 \exp(\frac{V}{80})$$

$$\alpha_m = \frac{0.1(V + 25)}{(\exp(\frac{V+25}{10}) - 1)}$$

$$\beta_m = 4 \exp(\frac{V}{18})$$

$$\alpha_h = 0.07 \exp(\frac{V}{20})$$

$$\beta_h = \frac{1}{(\exp(\frac{V+30}{10}) + 1)}$$

Equilibrium Potential:

$$E = \frac{RT}{zF} \ln \left(\frac{C_{\text{out}}}{C_{\text{in}}} \right)$$

Capacitance:

$$C = \frac{\epsilon \cdot A}{d}$$

Potassium Channel Activation:

$$n = n_\infty - (n_\infty - n_0) \exp\left(\frac{-t}{\tau_n}\right)$$

Sodium Channel Activation:

$$m = m_\infty - (m_\infty - m_0) \exp\left(\frac{-t}{\tau_m}\right)$$

Sodium Channel Inactivation:

$$h = h_\infty - (h_\infty - h_0) \exp\left(\frac{-t}{\tau_h}\right)$$

An Approximation: The Rinzel Hodgkin-Huxley Model

$$C \frac{dV}{dt} = -g_{\text{Na}} M(V)^3 (1 - R) (V - E_{\text{Na}}) - g_{\text{K}} R^4 (V - E_{\text{K}}) - g_{\text{leak}} (V - E_{\text{leak}}) + I$$

$$\frac{dR}{dt} = \frac{1}{\tau_R(V)} (-R + G(V))$$

Where:

- V: Neuronal membrane potential (in mV)
- I_{input} : Total input current (in μA)
- g_{K} : Conductance of K^+ ion channels (in mS/cm^2)
- g_{Na} : Conductance of Na^+ ion channels (in mS/cm^2)
- g_{leak} : Leakage conductance (in mS/cm^2)
- E_{K} : Equilibrium potential of K^+ ions (in mV)
- E_{Na} : Equilibrium potential of Na^+ ions (in mV)
- E_{leak} : Leakage equilibrium potential (in mV)
- C: Neuronal membrane capacitance (in $\mu\text{F}/\text{cm}^2$)
- R: Recovery variable, $R \in [0,1]$
- $M(V)$: Activation function for Na^+ ion channels, $M(V) \in [0,1] \forall V$
- $G(V)$: Activation function for the recovery variable R, $G(V) \in [0,1] \forall V$
- $\tau_R(V)$: Time constant for recovery variable R (in ms)

Auxiliary Equations for the Rinzel Hodgkin-Huxley Model

Voltage-Dependent Time Constant:

$$\tau_R(V) = 1 + 5 \exp\left(\frac{-(V+60)^2}{55^2}\right)$$

Equilibrium Potential:

$$E = \frac{RT}{zF} \ln\left(\frac{C_{\text{out}}}{C_{\text{in}}}\right)$$

Capacitance:

$$C = \frac{\varepsilon \cdot A}{d}$$

Conductance:

$$g = \frac{I}{(V - E)}$$

Steady State Values:

$$n_{\infty} = \frac{\alpha_n}{(\alpha_n + \beta_n)}$$

$$m_{\infty} = \frac{\alpha_m}{(\alpha_m + \beta_m)}$$

$$h_{\infty} = \frac{\alpha_h}{(\alpha_h + \beta_h)}$$

Resting State Values:

$$n_0 = \frac{\alpha_{n0}}{\alpha_{n0} + \beta_{n0}}$$

$$m_0 = \frac{\alpha_{m0}}{\alpha_{m0} + \beta_{m0}}$$

$$h_0 = \frac{\alpha_{h0}}{\alpha_{h0} + \beta_{h0}}$$

Slope of Spike Trajectory Projection:

$$S = \frac{1-h_0}{n_0}$$

Voltage-Dependent Transition Function:

$$M(V) = m_{\infty}$$

Recovery Variable Steady State:

$$G(V) = \frac{S(n_{\infty} + S(1-h_{\infty}))}{1+S^2}$$

Rate Constants:

$$\alpha_n = \frac{0.01(V+10)}{(\exp(\frac{V+10}{10}) - 1)}$$

$$\beta_n = 0.125 \exp(\frac{V}{80})$$

$$\alpha_m = \frac{0.1(V+25)}{(\exp(\frac{V+25}{10}) - 1)}$$

$$\beta_m = 4 \exp(\frac{V}{18})$$

$$\alpha_h = 0.07 \exp(\frac{V}{20})$$

$$\beta_h = \frac{1}{(\exp(\frac{V+30}{10}) + 1)}$$

Solution Analysis: Linearization + Eigenvalue Calculation Methods



Step 1:

Redefining the System

$$f_1(V, R) = \frac{dV}{dt} = \left[\frac{(-g_{Na}M(V)^3(1-R)(V-E_{Na}) - g_K R^4(V-E_K) - g_{leak}(V-E_{leak}) + I)}{C} \right]$$

$$f_2(V, R) = \frac{dR}{dt} = \left[\frac{1}{\tau_R(V)} (-R + G(V)) \right]$$

Step 2:

Finding the Equilibrium Points

$$f_1(V, R) = 0$$

$$f_2(V, R) = 0$$

$$V = V_{eq}$$

$$R = R_{eq}$$

Step 5:

Finding the Eigenvalues of the System

$$\det(\mathbf{J}_f(V_{eq}, R_{eq}) - \lambda I) = 0$$

Step 3:

Finding the Jacobian of the System

$$\mathbf{J}_f(V, R) = \begin{bmatrix} \frac{\partial f_1}{\partial V} & \frac{\partial f_1}{\partial R} \\ \frac{\partial f_2}{\partial V} & \frac{\partial f_2}{\partial R} \end{bmatrix} =$$

$$\begin{bmatrix} \left[\frac{(-3g_{Na}M(V)^2(M'(V))(V-E_{Na}-RV+RE_{Na})+(-g_{Na}M(V)^3)(1-R)-g_K R^4-g_{leak})}{C} \right] & \left[\frac{g_{Na}M(V)^3(V-E_{Na})-4g_K R^3(V-E_K)}{C} \right] \\ \left[\left(\frac{-\tau_R'(V)}{(\tau_R(V))^2} \right) (-R + G(V)) + \frac{1}{\tau_R(V)} (G'(V)) \right] & - \left[\frac{1}{\tau_R(V)} \right] \end{bmatrix}$$

Step 4:

Evaluating the Jacobian at the Equilibrium Points

$$\mathbf{J}_f(V_{eq}, R_{eq}) = \begin{bmatrix} \left[\frac{(-3g_{Na}(M(V_{eq}))^2(M'(V_{eq}))(V_{eq}-E_{Na}-R_{eq}V_{eq}+R_{eq}E_{Na})+(-g_{Na}M(V_{eq})^3)(1-R_{eq})-g_K R_{eq}^4-g_{leak})}{C} \right] & \left[\frac{g_{Na}M(V_{eq})^3(V_{eq}-E_{Na})-4g_K R_{eq}^3(V_{eq}-E_K)}{C} \right] \\ \left[\left(\frac{-\tau_R'(V_{eq})}{(\tau_R(V_{eq}))^2} \right) (-R_{eq} + G(V_{eq})) + \frac{1}{\tau_R(V_{eq})} (G'(V_{eq})) \right] & - \left[\frac{1}{\tau_R(V_{eq})} \right] \end{bmatrix}$$

Solution Analysis: *Trajectory Analysis*

- We can consider the behavior of the system around equilibrium points by way of *trajectories*; paths traced by the system's state as it evolves over time. Trajectories reveal information about system stability, limit cycles, and other dynamic behaviors.
 - Trajectory behavior is determined by the properties of eigenvalue pairs at equilibrium points.
 - We can use phase-plane diagrams to visualize trajectories.

Types of Equilibrium Points:

Spiral Point:

Eigenvalues are a complex conjugate pair. Asymptotically stable if the real part of the eigenvalues is negative, and unstable otherwise.

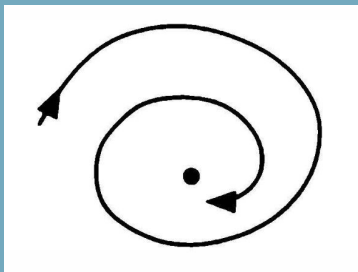


Figure 2: Diagram of a spiral point (Wilson, 2006).

Node Point:

Eigenvalues are both real, and have the same sign. Asymptotically stable if the eigenvalues are negative, and unstable otherwise.

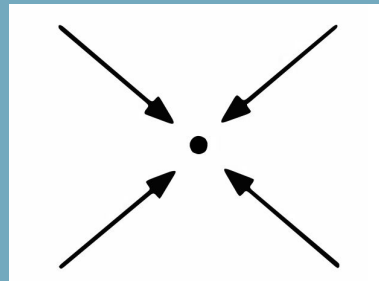


Figure 3: Diagram of a node point (Wilson, 2006).

Saddle Point:

Eigenvalues are both real, and have opposite signs. Saddle points are always unstable.

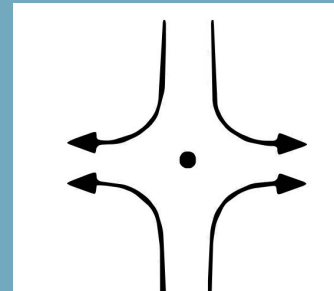


Figure 4: Diagram of a saddle point (Wilson, 2006).

Center:

Eigenvalues are pure imaginary. Centers are always stable, but not asymptotically.

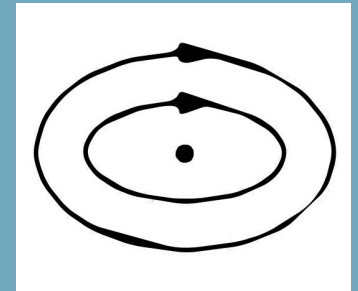


Figure 5: Diagram of a center (Wilson, 2006).

Solution Analysis: *Bifurcation Analysis*

- Nonlinear dynamical systems can exhibit *bifurcations*; events involving the appearance or disappearance of equilibrium points.
- The location at which bifurcations occur are known as *bifurcation points*.
- Bifurcation points provide insight into system stability and the existence of limit cycles.
- We can consider how equilibrium points change with varying system parameters via bifurcation analysis, to determine if and where bifurcation points appear.
- Bifurcation diagrams can be used to visualize equilibrium point changes.

Types of Bifurcations:

Hopf Bifurcations:

An equilibrium point loses stability as a complex conjugate pair crosses the imaginary axis. There are two kinds; Supercritical (an asymptotically stable limit cycle emerges as the resting state becomes unstable) and Subcritical (an unstable limit cycle merges with an asymptotically stable equilibrium state).

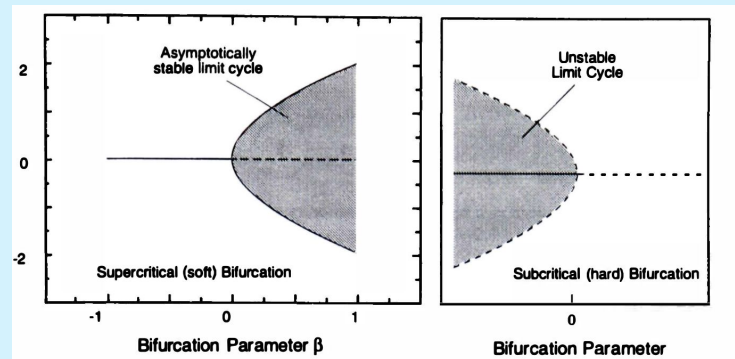


Figure 6: Hopf Bifurcations (Wilson, 2006).

Saddle-Node Bifurcations:

A saddle point and a node point coalesce and vanish. These bifurcations have the potential to produce limit cycles.

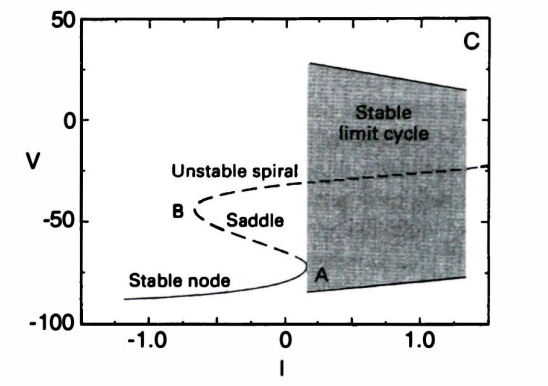


Figure 7: Saddle-Node Bifurcations (Wilson, 2006).

A Case Study: *The Clamped Squid Axon*

- We can consider the giant axon of the squid *Loligo forbesi*, which has been clamped to hold the membrane potential constant (at -70mV).
- The axon of *Loligo forbesi* is a suitable candidate for the Hodgkin-Huxley and Rinzel Hodgkin-Huxley models, as these models were initially developed to analyze firing rates in *Loligo forbesi*, which depend solely on Na^+ ion channels, K^+ ion channels, and leakage currents.
- Specifically, to model neuronal firings of this axon, we will use the Rinzel Hodgkin-Huxley model, alongside experimental data obtained from Hodgkin & Huxley (1952).

Modeling the Clamped Squid Axon: *Code Excerpts*

- We developed MATLAB code for *L. forbesi* neuron firing simulations.

```
% Trajectory Analysis for Loligo forbesi
Neuronal Firing Simulation

gNa = 120; % Sodium conductance (mS/cm^2)
gK = 36; % Potassium conductance (mS/cm^2)
gL = 0.3; % Leak conductance (mS/cm^2)
ENa = 55; % Sodium reversal potential (mV)
EK = -92; % Potassium reversal potential (mV)
EL = -50; % Leak reversal potential (mV)
C = 0.8; % Membrane capacitance (pF/cm^2)
```

```
% Code for finding equilibrium points and
plotting phase space diagram, with auxiliary
functions defined in other scripts
```

```
function [dV, dW] = system_derivatives(V, W)

m = minf(V);
w_inf = winf(V);
tau_w = tw(V);

INa = gNa * m^3 * (1-W) * (V - ENa);
IK = gK * W^4 * (V - EK);
IL = gL * (V - EL);

dV = (-INa - IK - IL + I) / C;
dW = (w_inf - W) / tau_w;

end

function dydt = system_ode(t, y)

V = y(1);
W = y(2);

[dV, dW] = system_derivatives(V, W);

dydt = [dV; dW];

end
```

```
function J = jacobian_matrix(V, W)

m = minf(V);

dm = exp(-(V + 35)/10) / (10 * (1 + exp(-(V + 35)/10))^2);

dV_dV = (-gNa * (3*m^2*dm*(1-W)*(V - ENa) + m^3*(1-W)) -
...
gK*W^4 - gL) / C;

dV_dW = (gNa*m^3*(V - ENa) - 4*gK*W^3*(V - EK)) / C;

dW_dV = exp(-(V + 60)/8) / (8 * (1 + exp(-(V + 60)/8))^2 *
...
(1 + 5*exp(-(V + 60)^2/55^2))) - ...
(winf(V) - W) * ...
(10*(V + 60)/55^2*exp(-(V + 60)^2/55^2)) / ...
(1 + 5*exp(-(V + 60)^2/55^2))^2;

dW_dW = -1/(1 + 5*exp(-(V + 60)^2/55^2));

J = [dV_dV dV_dW; dW_dV dW_dW];

end
```

Modeling the Clamped Squid Axon: Code Excerpts (Cont.)

```
% Initialize arrays for storing results
```

```
I_vals = -20:0.1:50;
```

```
results = struct( 'I', [], 'V', [], 'W', [], 'eigenvalues',  
[], 'stability', [], 'type', [], 'real_parts', [],  
'imag_parts', []);
```

```
idx = 1;
```

```
% Search for equilibrium points
```

```
for I_current = I_vals
```

```
I = I_current;
```

```
% Grid search for equilibrium points
```

```
V_range = -100:0.5:50;
```

```
% First pass: Find approximate equilibrium points
```

```
potential_equilibria = [];
```

```
for V = V_range
```

```
W = winf(V);
```

```
[dV, ~] = system_derivatives(V, W);
```

```
if abs(dV) < 1
```

```
potential_equilibria = [potential_equilibria; V];
```

```
end
```

```
end
```

```
% Cluster and refine equilibrium points
```

```
if ~isempty(potential_equilibria)
```

```
cluster_thresh = 2;
```

```
V_clusters = [];
```

```
i = 1;
```

```
while i <= length(potential_equilibria)
```

```
cluster = potential_equilibria(i);
```

```
potential_equilibria(i) = NaN;
```

```
nearby = abs(potential_equilibria - cluster) <  
cluster_thresh;
```

```
if any(nearby)
```

```
cluster = mean([cluster;  
potential_equilibria(nearby)]);
```

```
potential_equilibria(nearby) = NaN;
```

```
end
```

```
V_clusters = [V_clusters; cluster];
```

```
i = find(~isnan(potential_equilibria), 1);
```

```
if isempty(i)
```

```
break;
```

```
end
```

```
end
```

```
% Refine each cluster center
```

```
for V_approx = V_clusters'
```

```
try
```

```
F = @(x) system_derivatives(x(1), winf(x(1)));
```

```
[V_refined, ~, exitflag] = fsolve(@(x) F(x), V_approx,  
optimoptions( 'fsolve', 'Display', 'off'));
```

```
if exitflag > 0
```

```
V = V_refined;
```

```
W = winf(V);
```

```
[dV, dW] = system_derivatives(V, W);
```

```
if abs(dV) < 0.1 && abs(dW) < 0.1
```

```
J = jacobian_matrix(V, W);
```

```
eigs = eig(J);
```

```
stable = all(real(eigs) < 0);
```

```
% Store the real and imaginary parts of eigenvalues
```

```
results.real_parts{idx} = real(eigs);
```

```
results.imag_parts{idx} = imag(eigs);
```

```
if all(abs(imag(eigs)) < 1e-10)
```

```
if all(eigs > 0)
```

```
type = 'Unstable Node' ;
```

```
elseif all(eigs < 0)
```

```
type = 'Stable Node' ;
```

```
else
```

```
type = 'Saddle' ;
```

```
end
```

```
else
```

```
If stable
```

```
type = 'Stable Spiral' ;
```

```
else
```

```
type = 'Unstable Spiral' ;
```

```
end
```

```
end
```

```
results.I{idx} = I_current;
```

```
results.V{idx} = V;
```

```
results.W{idx} = W;
```

```
results.eigenvalues{idx} = eigs;
```

```
results.stability{idx} = stable;
```

```
results.type{idx} = type;
```

```
idx = idx + 1;
```

```
end
```

```
end
```

```
catch
```

```
continue;
```

```
end
```

```
end
```

```
end
```

```
end
```

Trajectory and Bifurcation Analysis

Phase Plane Diagram for $\frac{dV}{dt}$, $\frac{dR}{dt}$,
Linearized at Equilibrium Points

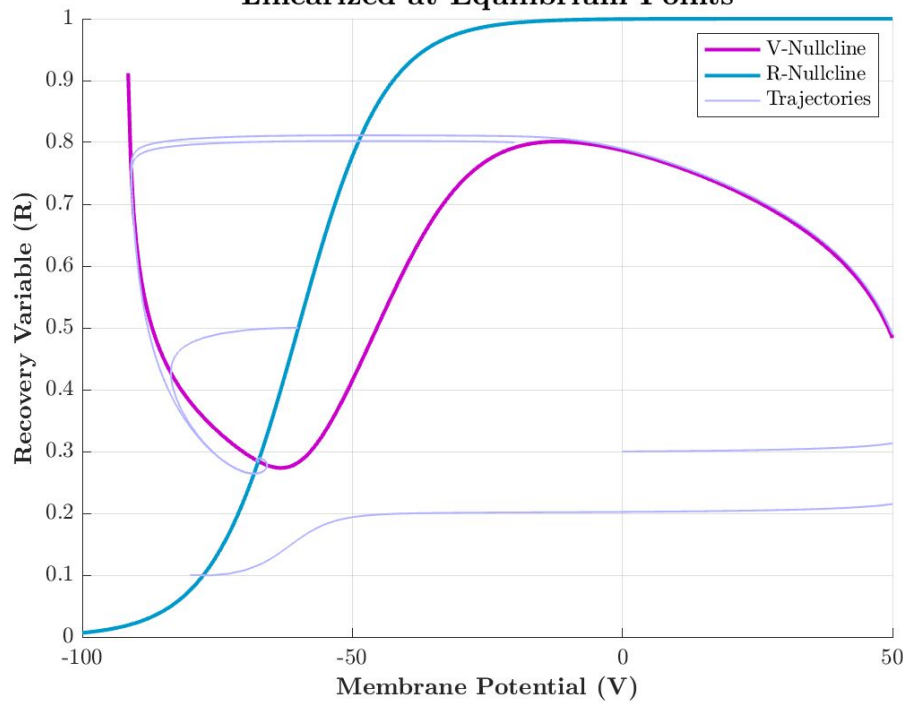
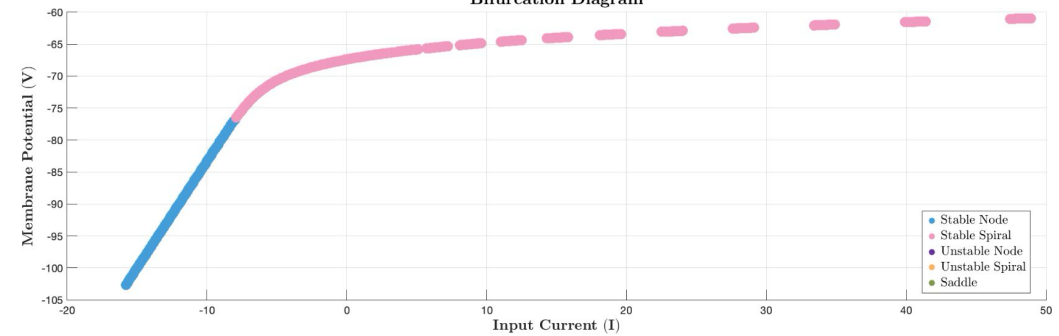


Figure 8: A Phase Plane Diagram for the Rinzel Hodgkin Huxley Model.

Bifurcation Diagram



Real Parts of Eigenvalues

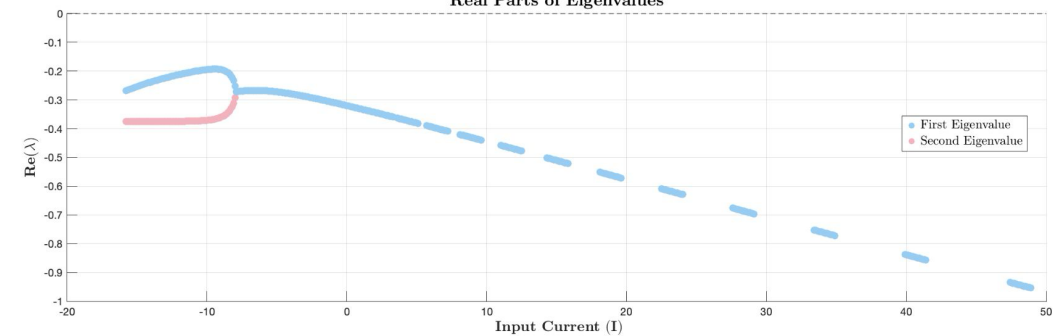


Figure 9: A Bifurcation Diagram with Two Subplots; the Top Subplot Relating V to I, and the Bottom Modeling Specific Eigenvalue Changes as I increases.

References

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